Curriculum Vitae

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Skills

- **Languages:** Python, Java, C/C++, MATLAB, R, C#
- **Frameworks/Tools:** PyTorch, Scikit-learn, Pandas, NumPy, Linux, Git, Docker
- **Expertise:** Software Engineering, Machine Learning, Data Science, Representation Learning, Manifold Learning, Graph Neural Networks, Time-Series Analysis, Optimization, Causal Inference

Professional Experience

DXT Commodities — Developer *2022 – 2025*

- Led the design and deployment of a centralized log aggregation system using the ELK (Elasticsearch, Logstash, Kibana) stack and Docker containers, significantly improving monitoring and debugging efficiency across internal services.
- Built crawlers and scrapers for internal log data reports from CRON jobs, enabling automated data collection and analysis for operational insights.
- Built WebSocket-based real-time data pipeline for high-frequency energy market data, enabling low-latency processing and analysis of electricity price fluctuations to filter out-of-distribution bids and offers in real-time using anomaly detection techniques.
- Built machine learning pipelines for forecasting electricity prices via supply/demand curve modeling using Time-Series Analysis.
- Contributed to the development and training of Generative AI systems (Transformer Models, etc.) for solar radiation forecasting from satellite imagery; outperformed ECMWF forecast models for solar radiation forecasts up to 4 hours ahead on MAE and RMSE.
- Developed predictive optimization framework for hydroelectric plant operations using shadow price modeling and long-term stochastic inflow forecasts using machine learning techniques such as Graph Neural Networks, along with convex optimization.

Università della Svizzera Italiana — Teaching Assistant *2024 – 2025*

- Taught and supported over 20 Bachelor students enrolled in Data Science, focusing on applied statistics and Data Analysis using Python and R.
- Designed programming exercises and tutorials to reinforce statistical modeling and data wrangling concepts.
- Led office hours and live sessions, helping students debug code, understand distributions, and interpret results from real-world datasets.
- Assisted instructors in grading and shaping the curriculum to enhance student engagement and learning outcomes.

Università della Svizzera Italiana — Computational Phylogenetics Research Intern *2022*

- Improved maximum likelihood estimation in diversification models by analyzing numerical instability in Monte Carlo importance sampling for phylogenetic tree likelihoods.
- Optimized evolutionary model parameters using Differential Evolution on CSCS supercomputing cluster; identified scalability bottlenecks and proposed machine learning alternatives.
- Prototyped a neural network pipeline for parameter inference from branching times, trained on simulated phylogenetic trees, and evaluated performance against traditional likelihood-based methods.
- Designed preprocessing and data simulation tools for generating phylogenetic trees and training datasets under speciation/extinction constraints.

Theses

- **Master's Thesis – Information Requirements and Allocation in Latent Representations:** Develops a rate-distortion and information-theoretic framework for optimal latent compression in representation learning, proving via the Data Processing Inequality that reconstruction is

the maximally information-demanding task and studies how β-weighted KL constraints and information bottlenecks govern the emergence and collapse of latent dimensions during training. In addition to which, defines a new disentanglement method that only allocates information in the latent dimensions relevant to the task, while suppressing information in nuisance dimensions.
- **Bachelor Thesis – Cryptocurrency Market Modelling Using Causal Inference Methods and Sentiment Analysis:** Developed a causal inference pipeline integrating VADER sentiment analysis with causal inference methods to evaluate the effect of social sentiment on crypto market movements, finding no significant causal effect of Twitter sentiment on crypto returns at standard significance levels.

Conference

- **Innovation in dynamic network modelling — Bernoulli Society for Mathematical Statistics and Probability:** Modelled patent citation dynamics using Relational Event Models (REMs) with sampled partial likelihood, incorporating endogenous network effects alongside time-varying textual and technological similarity covariates; identified content similarity and preferential attachment as the dominant drivers of citation propensity.

Manuscripts in Preparation

- **Relevance-Gated β-TCVAE**: Designs a training objective that decomposes the ELBO into relevance-aware components, applying sign-asymmetric per-dimension MI weighting to selectively reward information in relevant latent dimensions and suppress it in nuisance ones thereby constructing asymmetric behavioral incentives across a learned partition of the latent space — matches or beats state-of-the-art networks.

Education

- **MSc — Artificial Intelligence (Magna Cum Laude)** *2026*
  Università della Svizzera Italiana, Lugano
- **BSc — Informatics (Magna Cum Laude)** *2023*
  Università della Svizzera Italiana, Lugano

Projects

- **Spatiotemporal GNNs for Numerical Weather Prediction:** Designed and evaluated ST-GNNs to postprocess wind speed ensemble forecasts across 152 Swiss weather stations. Achieved 5% better CRPS score using a MultiScale Graph WaveNet, outperforming strong spatio-temporal baselines.
- **Swiss Insurance Chatbot:** Built a multilingual GenAI chatbot using retrieval-augmented generation (RAG), integrated with speech-to-text (Whisper) and text-to-speech (TTS) systems for full voice interaction. Tailored for regulatory Swiss insurance queries with modular evaluation and observability components.
- **Beta-Total Correlation VAE (TCVAE) Ablation Study:** Systematically studied how β-weighted KL objectives shape training dynamics, active dimension emergence, and posterior collapse across distortion regimes on dSprites; found sharp active/inactive KL separation matching known generative factors, demonstrating direct control of learned representation structure through objective weighting.
- **Beta-VAE Ablation Study:** Computed rate-distortion curves across latent dimensions $d \in \{3, \ldots, 8\}$ on dSprites; found $d \geq 5$ achieves near-optimal reconstruction while $d < 5$ incurs significant distortion penalties, empirically confirming convergence to intrinsic dimension of the data.
- **Relevance Factor VAE (RF-VAE) Ablation Study:** Confirmed that RF-VAE's learnable relevance vector converges to an effective gating mechanism on redundant latent dimensions, correctly identifying the informative subset on dSprites; this gating behavior directly motivates the relevance-aware weighting in the proposed Relevance-Aware β-TCVAE objective.

- **Multi-hop Influence Maximization:** Designed a GNN to approximate greedy influence (disease) spread in social networks using a custom fitness loss and GVAE-generated graphs (RMSE-close accuracy indicating network propagation).
- **CycleGAN Image Translation:** Trained a GAN-based model for unpaired style transfer; enforced cycle-consistency loss for domain alignment in visual mapping tasks.
- **Amazon Recommender System:** Built SVD- and neural embedding-based recommenders; achieved RMSE of 1.05 (SVD) and 1.45 (deep NN) on Amazon Video ratings.
- **Quora Duplicate Detection:** Fine-tuned BERT for semantic duplicate detection, achieving 83% accuracy — 5 points above a TF-IDF + Logistic Regression baseline; studied the gap between surface-level n-gram similarity and deep semantic representations.
- **Maze Solver Robot (group):** Built a ROS-based autonomous robot using Thymio in CoppeliaSim; implemented flood-fill path planning, visual servoing with OpenCV, and PID-aligned motion control.
- **DeveloperBuddy Gamification App (team lead):** Led a team of 10 in a 2.5-month full-stack software engineering project to build a gamified GitLab-integrated productivity app using Java (Spring Boot), Vue.js, and GitLab OAuth2. Designed and implemented challenge systems, badge mechanics, user analytics, and REST APIs. Contributed 23.2% of backend code, authored 18.3% of merged requests, and delivered over 9,400 lines of code. Achieved final grades of 97–120% across batches, including a 15% leadership bonus. Integrated CI workflows and backend testing pipelines using GitLab CI/CD for robust deployment and release management.
- **Spoople Athlete Search Engine:** Built a search engine for athlete biographies using Scrapy (data scraping), Apache Solr (indexing), and Vue.js (frontend). Parsed over 33k documents, implemented query highlighting, and evaluated usability via user studies, and containerized the application using Docker for scalable deployment.

Achievements

- **New York Academy of Sciences:** Invited to present a project on Food Fortification, analyzing nutrient deficiencies and suggesting food solutions without requiring a blood test (2017).
- **Infosys Spark - Catch Them Young Program:** Selected among top students across Jaipur, India for a two-week residential training at Infosys Campus focused on programming, problem-solving, and system design (2015).

Languages

English (fluent), Hindi (native), Spanish (intermediate), Italian (intermediate)